Analysis of Opportunistic Scheduling Algorithms in OFDMA Systems in the Presence of Generalized Fading Models

S. Kalyani and R. M. Karthik

***Abstract*—Analytical expressions for scheduling gain and spectral efficiency of the proportional fair and maximum rate scheduling algorithms for Orthogonal Frequency Division Multiple Access (OFDMA) based systems are derived for the following cases: a) multipath Rayleigh and multipath Nakagami fading, b) Composite channel models which model the combined effect of both small scale and large scale fading. It is shown using Extreme Value Theory (EVT) that the asymptotic distribution of the maxima of the received signal to noise power (SNR) across all the users converges to a Gumbel distribution for both cases. Therefore, we use the Gumbel distribution to derive expressions for both the spectral efficiency and scheduling gain. The scheduling gains obtained through numerical integration (whenever tractable) and simulations match with the analytical values obtained using the EVT based expressions. We also discuss how the moments and order statistics of the Gumbel distribution can be used to study other metrics of the scheduling algorithms.**

***Index Terms*—Composite fading models, scheduling algorithms, extreme value theory, Gumbel distribution, OFDM systems.**

I. Introduction and Related Work

In 4G cellular systems such as those based on 3$^{\rm rd}$ Generation Partnership Project (3GPP) Long Term Evolution (LTE) and LTE-Advanced (LTE-A) [1], the focus is on scheduling algorithms which *opportunistically* schedule users based on favorable channel conditions so as to achieve higher system throughput. Some popular opportunistic schedulers are the Max Rate Scheduler (MRS) and the Proportional Fair Scheduler (PFS). Various aspects of the PFS have been extensively studied (look in [2], [3], [4], [5] and the references therein) in the context of cellular systems. Throughput obtained with the MRS algorithm has been studied in [4], [6], and [7]. Extreme Value Theory (EVT) has been used to study bounds for throughput obtained with opportunistic scheduling algorithms in [6], [7], [8], and [9]. [6] analyses the asymptotic throughput of a system using max-sum-capacity (MSC) scheduling algorithm, and gives the asymptotic expressions for the system throughput applying techniques from EVT. In a similar vein, [7], [8] and [9] also apply EVT to analyze the throughput obtained with scheduling algorithms. However, most of the work has been focused on the scheduling gain of the opportunistic schedulers in the presence of single-tap small scale (Rayleigh or Nakagami-m) fading for single cell systems [3]-[9].

In this work, we will derive expressions for the scheduling gain and spectral efficiency achieved by PFS and MRS when compared with the Round Robin Scheduler (RRS) in OFDMA based single cell systems for the following cases: a) In the presence of multipath Rayleigh/Nakagami-m fading channel, b) In the presence of composite channel models. While one could attempt to compute the scheduling gain through numerical integration (using a software such as *Mathematica*), such an approach is not mathematically tractable when one considers moderate to large number of users (greater than ten). EVT will be used to show that the SNR of the user selected by PFS or MRS has a Gumbel probability density function (pdf) in the case of both small scale channel model and composite channel model. The Gumbel pdf is then used to derive the scheduling gains for both PFS and MRS. The scheduling gains computed theoretically using the Gumbel pdf match with values obtained through both simulation and numerical integration (whenever integration is tractable). Furthermore, the Gumbel pdf can also be used to compute various metrics such as variance of SNR of the user selected by the opportunistic scheduler, the median SNR of the selected user (50% of selected users see a SNR greater than the median SNR) in closed form.

II. System Model

We analyze the effect of the small scale and composite fading on the scheduling gain of PFS and MRS over RRS in a single cell system where there are n active users, each with an infinite buffer of data and no data rate constraints. Let $r_i(t)$ denote the transmission rate of user i at slot t if scheduled. It is assumed that $\{r_i(t), t \geq 0\}$ is a stationary and ergodic process. Similar to [3], it is assumed that rate prediction is perfect. Let $\tau_i(t)$ denote the received channel power of user i at time slot t on a subcarrier in the OFDM symbol. In OFDMA/OFDM systems, the SNR seen by the $i^{\rm th}$ user on a subcarrier at time slot t can be defined as $g\tau_i(t)$ where $g = \frac{E_s}{\sigma_T^2}$, E_s is the transmitted symbol energy and σ_T^2 is the thermal noise variance. MRS selects the user $i(t)$ at slot

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t according to

$$i(t) = \arg\max_{j=1,\dots,n} r_j(t).$$

PFS selects the user $i(t)$ at slot t with the highest transmission rate relative to its current average throughput, i.e.,

$$i(t) = \arg\max_{j=1,\dots,n} \frac{r_j(t)}{T_j(t)}. \tag{1}$$

The throughput $T_j(t)$ is typically evaluated through an exponentially smoothed average :

$$T_j(t) = \begin{cases} \left(1 - \frac{1}{t_c}\right) T_j(t-1) + \frac{1}{t_c} r_j(t), & j = i(t) \\ \left(1 - \frac{1}{t_c}\right) T_j(t-1), & j \neq i(t) \end{cases}$$

where t_c is the time-window length over which one wants to regulate the fairness. The PFS is asymptotically fair in the sense that all users receive the same fraction of the time slots in a homogeneous system [2], [3]. When $t_c \to \infty$, the average throughputs are stationary [3].

The spectral efficiency of RRS is

$$C_{RRS} = \frac{1}{n} \sum_{i=1}^{n} \int_0^\infty d(\tau_i) f(\tau_i) d\tau_i \tag{2}$$

where $f(\tau_i)$ is the pdf of τ_i. Here $d(\tau) = g\tau$ when rate is assumed to be linearly related to τ and $d(\tau) = \log_2(1 + g\tau)$ when the rate is assumed to be logarithmically related to τ. Similarly the spectral efficiency of MRS is given by [4]

$$C_{MRS} = \int_0^\infty d(\tau) f_{\max}(\tau) d\tau. \tag{3}$$

Here, $f_{\max}(\tau)$ is the pdf of $\max(\tau_1(t), ..., \tau_n(t))$, and $\max(\cdot)$ denotes the maximum element of $(\cdot)$. The SNR seen by the selected user is $g \max(\tau_1(t), ..., \tau_n(t))$. The spectral efficiency of PFS given the combined path loss and shadow loss component s_i of the i^{th} user is [4]

$$C_{PFS} = \frac{1}{n} \sum_{i=1}^{n} \int_0^\infty d(s_i \tau_t) \tilde{f}(\tau_t) d\tau_t. \tag{4}$$

The s_i-s for each user can be different (depends on the pathloss and shadow loss values and hence on the propagation scenario), and can vary with user drop [1]. However in a user drop, the s_i-s are treated as constant w.r.t. the time index t assuming that the users are not mobile, and this approach has been widely used in the analysis of PFS [4], [5], [10]. Here, $\tilde{f}(\tau)$ is the pdf of $\max(\tau_{t,1}, ..., \tau_{t,n})$ where $\tau_{t,i}$ is the time varying component of $\tau_i(t)$ (here it is the small scale fading component of the received channel) which is i.i.d. across users. In the case of small scale fading, $s_i = 1, \ \forall i$ and $\tau_i(t) = \tau_{t,i}$ is i.i.d. with $E[\tau_i(t)] = 1, \ \forall i$. In such a scenario as $t_c \to \infty$, both PFS and MRS are asymptotically equivalent [11]. In the case of composite fading, to obtain the average value of spectral efficiency of PFS one needs to take the expectation of (4) w.r.t. the pdf of $s_i, \ \forall i$.

[1]The term user drop refers to ways the users are dropped in a cellular network, i.e. their positions in the cell. As a standard practice, the users are uniformly dropped in the cell, i.e., the average number of users/unit area is constant within a cell.

A. The pdf of $\tau_i(t)$

The variations in the mobile wireless channel can be roughly divided into two types: 1) large scale fading, due to path loss of signal as a function of distance and shadowing by large objects, and 2) small scale fading, due to constructive and destructive interference of the multiple signal paths between the transmitter and receiver. Typically, small-scale fading is modelled using Ricean or Nakagami-m distributions, while large scale shadow fading is modelled using the lognormal distribution. For more details on large scale and small scale fading, please refer to [12]. In wireless channels, both small scale fading and large scale fading take place simultaneously leading to the phenomenon referred to as composite fading [13].

1) Small scale fading: The wireless channel $\mathbf{h}_i(t)$ in time domain at slot t of the i^{th} user comprises of L taps and has unit energy. It has been shown in [14] that for OFDM systems with multipath Nakagami-m/Rayleigh fading, the received channel power $\tau_i(t)$ seen on a subcarrier for the i^{th} user has a gamma pdf given by

$$f_{\tau_i(t)}(x) = \frac{x^{k-1} e^{-\frac{x}{\theta}}}{\theta^k \Gamma(k)} \tag{5}$$

where $\Gamma(\cdot)$ is the Gamma function. The shape parameter k and scale parameter θ are given by

$$k = \frac{\left(\sum_{j=0}^{L-1} a_j\right)^2}{\sum_{j=0}^{L-1} \frac{a_j^2}{m_j} + \sum_{l=0}^{L-1} \sum_{\substack{j=0,\\ j\neq l}}^{L-1} a_l a_j}, \tag{6}$$

$$\theta = \frac{\sum_{j=0}^{L-1} a_j}{k} \tag{7}$$

where a_j is the variance of the j^{th} time domain channel tap of the i^{th} user, i.e., $E[h_{ij}(t) h_{ij}^*(t)] = a_j$ and m_j represents the corresponding fading severity and is a parameter of the Nakagami-m distribution. Since small scale fading is i.i.d. across users, m_j-s and a_j-s are same for all users. Furthermore, $\sum_{j=0}^{L-1} a_j = 1$ since the time domain channel $\mathbf{h}_i(t)$ is assumed to have unit energy. In the case of multipath Rayleigh fading, $k = \theta = 1$ since $m_j = 1 \ \forall j$ and hence the $\tau_i(t)$ are independent and exponentially distributed. .

2) Composite channel model: The pdf of the received channel power $\tau_i(t)$, conditioned on the fact that the average channel power is Ω is given by [13]:

$$f_{\tau_i(t)|\Omega}(\tau) = \frac{1}{\Gamma(k)} \left(\frac{k}{\Omega}\right)^k \tau^{k-1} \exp\left(-\frac{k\tau}{\Omega}\right), \quad \tau \geq 0, \quad k \geq 0.5. \tag{8}$$

The variability associated with large-scale environmental obstacles leads to the local mean power Ω fluctuating about a mean power P_r. This phenomenon is known as shadowing [15]. If a variable gain amplifier can compensate for the variations in Ω, then the pdf of $\tau_i(t)$ can still be given by (5) with $\Omega = k\theta = 1$.

Though empirical studies indicate that Ω has a lognormal pdf, it has been shown widely [13], [15] that using lognormal large scale fading model will not result in closed-form expressions for the pdf of the received signal power in the case of the composite channel. It has been suggested in [13], [15] that the channel power due to shadowing for the i^{th} user can be approximated by Gamma distribution with shape parameter m_s and scale parameter $\frac{\Omega_i}{m_s}$. Using moment matching between the moments of lognormal and gamma pdf, [15] shows that

$$m_s = 1/\left(\exp\left(\frac{\sigma}{8.686}\right)^2 - 1\right) \quad (9)$$

$$\Omega_i = P_{r,i}\sqrt{(m_s+1)/m_s} \quad (10)$$

where σ is the standard deviation of the lognormal pdf in dB and $P_{r,i}$ is the constant area mean power (in a small area) seen in the vicinity of the i^{th} user. Here, the expected value of the shadowing plus path loss component is given by $E[s_i] = \Omega_i$. The pdf of $\tau_i(t)$ in the case of the composite model is then given by the generalized-K pdf [13], [15]

$$f_{\tau_i(t)}(\tau) = \frac{2}{\Gamma(k)\Gamma(m_s)}\left(\frac{l_i}{2}\right)^{k+m_s}\tau^{\left(\frac{k+m_s}{2}\right)-1}K_{m_s-k}(l_i\sqrt{\tau}), \quad (11)$$

where $\tau > 0$, $k > 0.5$ and $m_s > 0$. Here, $K_{m_s-k}(\cdot)$ is the modified Bessel function of the second kind of order $(m_s - k)$ and $l_i = 2\sqrt{\frac{km_s}{\Omega_i}}$. The mean value of $\tau_i(t)$ is given by $E[\tau_i(t)] = \Omega_i$ [13]. Note that the pdfs of the users are non-identical since Ω_i is a function of the user location. The corresponding cdf is given by (12), where ${}_1F_2(a; b, c; x)$ is the generalized hypergeometric function.

B. Evaluation of scheduling gain of PFS and MRS over Round-Robin scheduler

The scheduling gain of a scheduler w.r.t. RRS is defined as the ratio of throughput (spectral efficiency) achieved under that scheduler compared to the throughput (spectral efficiency) under RRS. Here, $G(n)$ denotes the scheduling gain when $d(\tau) = g\tau$ (linear relationship between the rate $r(t)$ and SNR) and $\tilde{G}(n)$ denotes the scheduling gain when $d(\tau) = \log_2(1 + g\tau)$ (logarithmic relation between rate and SNR).

1) Scheduling gain when $d(\tau) = g\tau$: The scheduling gain, $G(n)$ in the presence of n users in case of small scale fading wherein the τ_i are i.i.d. gamma variates for both PFS and MRS is

$$\begin{aligned} G_{PFS,S}(n) = G_{MRS,S}(n) &= \frac{\int_0^\infty \frac{nx^k e^{-\frac{x}{\theta}}}{\theta^k\Gamma(k)}\left(\frac{\gamma(k,x/\theta)}{\Gamma(k)}\right)^{n-1}dx}{k\theta} \\ &= \int_0^\infty 1 - \left(\frac{\gamma(k,x/\theta)}{\Gamma(k)}\right)^n dx \end{aligned} \quad (13)$$

where $k\theta = 1$ as $\sum_{j=1}^{L} a_j = 1$ and $\left(\frac{\gamma(k,x/\theta)}{\Gamma(k)}\right)^n$ is the cumulative distribution function (cdf) of the maxima of the n i.i.d. gamma variates. Here, $\gamma(s,x)$ is the lower incomplete Gamma function [16] and the subscript S indicates that only small scale fading is being considered. In the case of composite channel model, the scheduling gain $G(n)$ of the MRS is obtained by evaluating

$$\begin{aligned} G_{MRS,C}(n) &= \frac{E[\max(\tau_1(t), ..., \tau_n(t))]}{E[\frac{1}{n}\sum_{i=1}^n \tau_i(t)]} \\ &= \frac{\int_0^\infty (1 - \prod_{i=1}^n F_{\tau_i(t)}(x))dx}{\frac{1}{n}\sum_{i=1}^n \Omega_i} \end{aligned} \quad (14)$$

where $F_{\tau_i(t)}(x)$ is the cdf given in (12). The subscript C indicates that the composite fading model is being considered. Taking expectation of (4) w.r.t. s_i to obtain the average spectral efficiency of PFS and using (2), $G_{PFS,C}(n)$ is given by

$$G_{PFS,C}(n) = \frac{\frac{1}{n}\sum_{i=1}^n E_{s_i}\left[\int_0^\infty s_i\tau_t\tilde{f}(\tau_t)d\tau_t\right]}{\frac{1}{n}\sum_{i=1}^n \int_0^\infty \tau_i f(\tau_i)d\tau_i}. \quad (15)$$

where $E_{s_i}[x]$ denotes the expectation of x w.r.t. s_i. Simplifying and using the fact that the random variables s_i and $\tau_t = \max(\tau_{t,1}, ..., \tau_{t,n})$ are independent and the small scale fading is i.i.d., one obtains [10]

$$\begin{aligned} G_{PFS,C}(n) &= \frac{\frac{1}{n}\sum_{i=1}^n E[s_i]\int_0^\infty 1 - \left(\frac{\gamma(k,x/\theta)}{\Gamma(k)}\right)^n dx}{\frac{1}{n}\sum_{i=1}^n \Omega_i} \\ &= \int_0^\infty 1 - \left(\frac{\gamma(k,x/\theta)}{\Gamma(k)}\right)^n dx. \end{aligned} \quad (16)$$

Hence, the scheduling gain $G_{PFS,C}(n)$ is only a function of small scale fading and is equal to $G_{PFS,S}(n)$. Using a slightly different argument, [5] also shows that $G_{PFS,C}(n) = G_{PFS,S}(n)$.

2) The scheduling gain when $d(\tau) = \log_2(1 + g\tau)$: Since the assumption of a linear rate function is not strictly true with increasing SNR, in this section we will give expressions for the scheduling gains of the PFS and MRS schedulers for the case where the rate is a logarithmic function of SNR. The scheduling gain achieved by PFS and MRS under small scale fading conditions when the rate and SNR are logarithmically related is given by

$$\begin{aligned} &\tilde{G}_{PFS,S}(n) = \tilde{G}_{MRS,S}(n) \\ &= \frac{n\int_0^\infty \log_2(1+gx)\left(\gamma(k,x/\theta)\right)^{(n-1)} x^{k-1}e^{-\frac{x}{\theta}}dx}{(\Gamma(k))^{n-1}\int_0^\infty \log_2(1+gx)x^{k-1}e^{-\frac{x}{\theta}}dx}. \end{aligned} \quad (17)$$

For the case when rate has a logarithmic relation to SNR, simulation results indicate that the scheduling gains $\tilde{G}_{PFS,C}(n)$ and $\tilde{G}_{PFS,S}(n)$ are numerically very close though to the best of our knowledge there is no theoretical analysis to show that $\tilde{G}_{PFS,C}(n) = \tilde{G}_{PFS,S}(n)$.

The scheduling gain of MRS in the presence of the composite channel is given by

$$\tilde{G}_{MRS,C}(n) = \frac{\int_0^\infty \log_2(1+gx)\frac{d}{dx}(\prod_{i=1}^n F_{\tau_i(t)}(x))dx}{\frac{1}{n}\sum_{i=1}^n\int_0^\infty \log_2(1+gx)f_{\tau_i(t)}(x)dx} \quad (18)$$

where $f_{\tau_i(t)}(x)$ is the generalized-K pdf given by (11) and $F_{\tau_i(t)}(x)$ is the corresponding cdf given by (12).

The expressions given above for scheduling gains can only be evaluated numerically using *Mathematica* and even that becomes increasingly cumbersome and/or mathematically intractable as n increases. The only case when the scheduling gain $G(n)$ can be computed exactly is the case of single-tap Rayleigh fading [3] where $G_{PFS,S}(n) = G_{MRS,S}(n) = \sum_{k=1}^n \frac{1}{k}$. For deriving expressions for the scheduling gain and computing the spectral efficiency of the schedulers, we require the pdf of maximum SNR given the n users' SNRs and this pdf is highly complicated in the case of Nakagami-m fading and composite fading models. We use EVT to show that the

$$F_{\tau_i(t)}(\tau) = \frac{2}{\Gamma(k)\Gamma(m_s)}\pi csc(\pi(m_s-k))\Bigg[\left(\frac{l_i}{2}\right)^k \Gamma(k)\tau^k {}_1F_2(k;1-m_s+k;k+1;l_i^2\tau) - \left(\frac{l_i}{2}\right)^{m_s}\Gamma(m_s)\tau^{m_s} {}_1F_2(m_s;1+m_s-k;m_s+1;l_i^2\tau)\Bigg] \quad (12)$$

asymptotic pdf of $\max(\tau_1(t), ..., \tau_n(t))$ is the Gumbel pdf for both small scale fading model and composite fading model even though the $\tau_i(t)$-s are not identically distributed in the case of composite fading. The Gumbel distribution is then used to derive expressions for $G(n)$ and $\tilde{G}(n)$ for both PFS and MRS.

III. Extreme Value Theory Based Analysis of Scheduling Algorithms

EVT is usually applied to study the behavior of the maxima of n i.i.d. random variables, and has found extensive use in areas such as finance, economics, weather forecasting, signal processing and the like. Though EVT deals with the asymptotic pdf of the maxima, it has been seen to hold even for n as small as ten or twenty [17]. The usefulness of EVT is because of the fundamental theorem given below:

Theorem 3.1: Fisher-Tippet Theorem, Limit Laws for Maxima [18]:
Let $X_1, X_2, \ldots, X_n$ be a sequence of n i.i.d. random variables and $M_n = \max(X_1, X_2, \ldots, X_n)$. If there exist constants $a_n > 0$ and $b_n \in \mathbb{R}$ and some non degenerate cdf $\mathcal{H}$ such that as $n \to \infty$

$$a_n^{-1}(M_n - b_n) \xrightarrow{d} \mathcal{H} \quad (19)$$

where $\xrightarrow{d}$ denotes convergence in distribution, then the cdf of $\mathcal{H}$ is one of the three cdfs:
Gumbel: $\phi_1(x) = \exp(-e^{-x})$, $x \epsilon \mathbb{R}$
Fréchet: $\phi_F(x) = \begin{cases} 0, & x \le 0 \\ e^{-x^{-\beta}}, & x > 0 \end{cases} \quad \beta > 0$
Weibull: $\phi_W(x) = \begin{cases} e^{-(-x)^{\beta}}, & x \le 0 \\ 1, & x > 0 \end{cases} \quad \beta > 0.$

Proof: The details of the proof can be found in [18]. ∎

Definition: Maximum Domain of Attraction [18] : We say that the cdf F of the random variable X belongs to the maximum domain of attraction ($\mathcal{MDA}$) of the extreme value distribution (EVD) $\mathcal{H}$ if there exist constants $a_n > 0$ and b_n $\in \mathbb{R}$ such that (19) holds.

Proposition: Suppose the cdf F is a von Mises function then it belongs to the $\mathcal{MDA}$ of the Gumbel distribution (ϕ_1) [18] with a possible choice of normalizing constants being

$$b_n = F^{-1}(1-\frac{1}{n}); \quad a_n = \frac{1-F(b_n)}{f(b_n)}. \quad (20)$$

f is the pdf corresponding to the cdf F and $F^{-1}(x) = \inf\{z : F(x) \ge z\}$. Since the $\tau_i(t)$ values are always non-negative, b_n in Equation (19) is positive. Let f' be the derivative of f. F is a von Mises function if

$$\lim_{x\to\infty} \frac{f'(x)[1-F(x)]}{f^2(x)} = -1. \quad (21)$$

A. *The pdf of* $\max(\tau_1(t), ..., \tau_n(t))$ *in the case of small scale fading*

Since $\tau_i(t)$ is a gamma distributed random variable, the derivative of its pdf, $f'(x)$ is

$$f'(x) = \frac{(k-1)x^{k-2}e^{-\frac{x}{\theta}}}{\theta^k\Gamma(k)} - \frac{x^{k-1}e^{-\frac{x}{\theta}}}{\theta^{k+1}\Gamma(k)} = f(x)(\frac{k-1}{x} - \frac{1}{\theta}). \quad (22)$$

Therefore,

$$\frac{f'(x)(1-F(x))}{f^2(x)} = \frac{(k-1)(1-F(x))}{xf(x)} - \frac{1-F(x)}{\theta f(x)} \quad (23)$$

which tends to the form $\frac{0}{0}$ as $x \to \infty$. Applying L' Hospital's rule and differentiating the numerator $(1 - F(x))$ and denominator $(f^2(x)/f'(x))$ separately one obtains

$$\lim_{x\to\infty} \frac{f'(x)(1-F(x))}{f^2(x)} = \lim_{x\to\infty} -1 + \frac{(k-1)\theta^2}{(k\theta - x)((k-1)\theta - x)} = -1. \quad (24)$$

Hence, the gamma distribution is a von Mises function and lies in the $\mathcal{MDA}$ of Gumbel distribution. Therefore, $\tau = \max(\tau_1(t), ..., \tau_n(t))$ is Gumbel distributed with cdf

$$\phi_{G,\tau}(x) = \exp[-e^{-\frac{x-b_n}{a_n}}]. \quad (25)$$

Here a_n and b_n are calculated using (20) where F is the gamma cdf.

B. *The pdf of* $\max(\tau_1(t), ..., \tau_n(t))$ *in the presence of composite fading channel*

EVT is used to find the pdf of $\max(\tau_1(t), ..., \tau_n(t))$ where the τ_i are independent and non-identically distributed (i.n.i.d.) generalized-K variables. To show that the asymptotic pdf of $\max(\tau_1(t), ..., \tau_n(t))$ is the Gumbel distribution in the case of composite fading, we need to first show that the cdf of $\tau_i(t)$ is a von Mises function.

Theorem 3.2: The generalized-K cdf is a von Mises function.

Proof: See Appendix A. ∎

Therefore, the distribution of the maxima of n i.i.d. generalized-K distributed variables converges in distribution to a Gumbel distribution. In general, the pdf of $\tau_i(t)$ depends on the user's location and the mean local area power P_r seen by the user. Therefore the pdf of $\tau_i(t)$ is not i.i.d. across users in the case of composite fading model. Instead the pdfs are independent but non-identical.

1) The pdf of maxima of i.n.i.d. generalized-K variates: The pdfs corresponding to the n users differ only due to their mean values Ω_i-s. In other words, $\tau_1(t), \tau_2(t) - (\Omega_2 - \Omega_1), ..., \tau_n(t) - (\Omega_n - \Omega_1)$ are i.i.d. random variables with the distribution given by the generalized-K distribution corresponding to the user with received power $\tau_1(t)$. EVT can still be applied to find the maxima of independent but non-identical generalized-K pdfs as these pdfs differ only due to varying mean values [19].

Consider n i.i.d. random variables $Z_1, ...Z_n$ wherein the normalized maxima M_n is in the $\mathcal{MDA}$ of the Gumbel distribution with normalizing constants b_n and $a_n > 0$. Define a sequence of point process P_n on $\mathbb{R}_+ \times \mathbb{R}$ with points $\{(\frac{k}{n}, \frac{Z_k - b_n}{a_n}), k = 1, 2, \ldots\}$. The following theorem from [20] gives the weak convergence of the point process to Poisson process.

Theorem 3.3: Let $\{Z_k, k \geq 1\}$ be i.i.d. random variables which satisfy (19), and suppose P_n is the point process on $\mathbb{R}_+ \times \mathbb{R}$ with points $\{(\frac{k}{n}, \frac{Z_k - b_n}{a_n}), k \geq 1\}$. Then there exist point processes U and U_k, $k \geq 1$ such that

1) $P_n =_d U_n$ for $n \geq 1$
2) U is Poisson random measure whose mean measure evaluated at $[0, t] \times (x, \infty)$ is $-t \log \mathcal{H}(x)$.
3) With probability one, $U_n \to U$.

Proof: Refer to Theorem 1 in [20]. ■

Here, $=_d$ denotes weak convergence of a point process to a random measure. We consider the case where the variables Z_k lie in the $\mathcal{MDA}$ of the Gumbel distribution. If $\{T_k, X_k\}_{k=1}^{\infty}$ is the enumeration of the points of the above Poisson point process, then we have the following theorem from [19].

Theorem 3.4: For $c > 0$ the sequence of stochastic processes

$$\left\{ \frac{\max_{0 \leq k \leq nt}(Z_k + cb_k) - (c+1)b_n}{a_n} \right\}_{t>0}$$

converges in $D(0, \infty)$ to the stochastic process

$$\{M_c(t)\}_{t>0} := \left\{ \sup_{0 \leq T_k \leq t} (X_k + c \log T_k) \right\}_{t>0}.$$

Proof: Refer to Theorem 2.1 in [19]. ■

Here $D(0, \infty)$ stands for the space of right-continuous functions on $(0, \infty)$ with left-hand limits [21]. The properties of the process $\{M_c(t)\}$ can be obtained from that of $\{M_0(t)\}$ [22]. The following theorem from [19] gives the relationship between the two processes.

Theorem 3.5:

$$\{M_c(t)\}_{t>0} \stackrel{d}{=} \left\{ M_0 \left(\frac{t^{c+1}}{c+1} \right) \right\}_{t>0}.$$

Proof: Refer to Theorem 2.2 in [19]. ■

Using Theorem 3.5 it can be shown that [19], [22]

$$P \left\{ \max_{1 \leq k \leq n} (Z_k + cb_k) - (c+1)b_n \leq a_n x \right\} = e^{-(c+1)^{-1} e^{-x}}. \tag{26}$$

Writing τ_i as $Z_i + cb_i$ with an appropriately chosen value for c, the cdf of $\max(\tau_1, ..., \tau_n)$ is given by $e^{-(c+1)^{-1} e^{(-\frac{x-(c+1)b_n}{a_n})}}$. The corresponding pdf is given by

$$f_{\max(\tau_1,...,\tau_n)}(x) = \frac{e^{(-\frac{x-(c+1)b_n}{a_n})}}{a_n(c+1)} e^{(-(c+1)^{-1} e^{(-\frac{x-(c+1)b_n}{a_n})})}. \tag{27}$$

This can be rewritten as a Gumbel pdf of the form

$$f_{\max(\tau_1,...,\tau_n)}(x) = \frac{e^{-\frac{x-(c+1)b_n + \log(c+1)a_n}{a_n}}}{a_n} e^{-e^{(-\frac{x-(c+1)b_n + \log(c+1)a_n}{a_n})}}. \tag{28}$$

Specific to our application, the pdf of Z_i corresponds to that generalized-K pdf among the n non-identical pdfs which has the lowest mean value, i.e. the pdf with $\Omega_i = \Omega_{\min}$. Here c is the chosen to be the value which best fits the set of equations

$$cb_i = \Omega_i - \Omega_{\min} \ \ \forall i \tag{29}$$

where $b_i = F^{-1}(1 - \frac{1}{i})$ (similar to the definition of b_n in (20)) and F is the generalized-K cdf with the mean value $\Omega_{\min}$. For example, c can be computed as the mean or trimmed mean of $\frac{\Omega_i - \Omega_{\min}}{b_i}, \ i = 1, ..., n$.

C. The EVT based scheduling gain when rate is a linear function of $\tau_i(t)$

From section II-B1, it is apparent that the scheduling gain $G(n)$ is the mean value of the maxima of the n $\tau_i(t)$ values. If the cdf of a random variable is in the $\mathcal{MDA}$ of an EVD, then from [6], [23], one can infer that the moments of the maxima of the n random variables converge to the moments of the limiting EVD if the random variables are non-negative. Since the support of both the gamma pdf and the generalized-K pdfs is non-negative, the moments of the maxima of n i.i.d. gamma variates and n i.n.i.d. generalized-K variates converge to the moments of their corresponding EVDs.

Since the gamma distribution is in the $\mathcal{MDA}$ of the Gumbel distribution, therefore $G_{PFS,S}(n)$ and $G_{MRS,S}(n)$ given in (13) are equal to the mean of the corresponding Gumbel distribution. The mean of the Gumbel distribution is $b_n + \gamma a_n$, where $\gamma = 0.5772$ is the Euler-Mascheroni constant. In the case of small scale fading, we have compared these scheduling gain values computed using the Gumbel distribution with the scheduling gain value obtained by numerically integrating (13), and the values are similar even for values of n as small as 10 and 20.

The generalized-K pdf is a von Mises function and the pdf of maxima of n i.n.i.d. generalized-K random variables is given by the Gumbel pdf in (28). Therefore, the mean value of the maxima is given by $a_n\gamma - a_n \log(c+1) + (c+1)b_n$. Hence the scheduling gain $G_{MRS,C}(n)$ is given by

$$G_{MRS,C}(n) = \frac{a_n\gamma - a_n \log(c+1) + (c+1)b_n}{\frac{1}{n}\sum_{i=1}^{n} \Omega_i}. \tag{30}$$

It is obvious from (30) that when $c = 0$, the numerator simply corresponds to the mean of the Gumbel pdf with parameters a_n and b_n. When c increases, the scheduling gain also increases since the term $(c+1)$ increases faster than that of $\log(c+1)$. Furthermore, it is also clear that $G_{MRS,C}(n)$ is a function of c and can vary from one user drop to another since c itself is a function of user drop.

While computing $G(n)$ using (13), (14) and (15) can be cumbersome and in some cases mathematically intractable, the Gumbel pdf based computation of $G(n)$ is very easy. For example, the scheduling gain computation using (30) is trivially simple in comparison to Equation (14) which involves the integral of the product of n non-identical cdfs wherein the cdf itself involves generalized hypergeometric functions.

D. Scheduling gain computation when $d(\tau) = \log_2(1 + g\tau)$ using the Gumbel pdf

For the case where the rate is a logarithmic function of $\tau_i(t)$, we evaluate (3) and (4) with $d(\tau) = \log_2(1 + g\tau)$ using the Gumbel pdf as the pdf of $\max(\tau_1(t), ..., \tau_n(t))$. Substituting for $f_{\max}(x)$ in (3) a Gumbel pdf with parameters a_n and b_n, we obtain the spectral efficiency as

$$C_{MRS} = \frac{1}{a_n \log 2} \int_0^{\infty} \log(1 + gx) e^{(-\frac{x-b_n}{a_n})} e^{(-e^{-\frac{x-b_n}{a_n}})} dx. \quad (31)$$

Equation (31) cannot be evaluated analytically even using a software such as *Mathematica*. Therefore, we derive a series expression for Equation (31) which is given by

$$C_{MRS} = \frac{1}{\log 2} \sum_{m=0}^{\infty} \frac{(-1)^m}{(m+1)!} e^{\frac{(m+1)(1+gb_n)}{ga_n}} \Gamma(0, \frac{m+1}{ga_n}). \quad (32)$$

For the derivation, please refer to Appendix B.

The spectral efficiency of PFS can be computed in a similar manner for the small scale fading model. Furthermore, since the pdf of $\max(\tau_1(t), ..., \tau_n(t))$ is a Gumbel pdf even in the case of the composite channel model, $C_{MRS,C}(n)$ can also be computed in a similar fashion.

The scheduling gain of PFS in the case of small scale fading assuming logarithmic relation relationship between rate and SNR is given by

$$\tilde{G}_{PFS,S}(n) = \frac{\sum_{m=0}^{\infty} \frac{(-1)^m}{(m+1)!} e^{\frac{(m+1)(1+gb_n)}{ga_n}} \Gamma(0, \frac{m+1}{ga_n})}{\int_0^{\infty} \log(1 + g\tau) \frac{\tau^{k-1} e^{-\frac{\tau}{\theta}}}{\theta^k \Gamma(k)} d\tau}. \quad (33)$$

$\tilde{G}_{MRS,C}(n)$ can be computed in a similar fashion. Though the $\tilde{G}(n)$ expressions involve an infinite series expression, as the series is converging (see Appendix B for details), the spectral efficiency can be computed using only a finite number of terms. Furthermore, the $\tilde{G}(n)$ expressions given here are much less complicated than (17) and (18).

E. Additional metrics that can be computed using the Gumbel distribution

In addition to scheduling gain, we can also obtain other parameters which can be used to compare different scheduling algorithms. The Gumbel distribution is the cdf of the received channel power τ of the user selected by PFS and MRS at each scheduling instant. Hence metrics such as variance of τ, percentage of selected users with channel power values exceeding a certain threshold, etc., can be studied by looking at the moments and order statistics of the Gumbel distribution. For example, $b_n - a_n \log\log n$ is the median of the Gumbel distribution (median channel power) and hence specific to our application, 50% of the scheduled users in a time period see a SNR greater than $g(b_n - a_n \log\log n)$. As the number of users increases, the median SNR increases. Therefore, from the median SNR value we can compute the number of users required for 50% of the scheduled users to have a SNR greater than a specific threshold value. The variance of the Gumbel distribution is given by $\frac{\pi^2}{6} a_n^2$ and it provides information on how the SNR of the selected user varies in every scheduling instant. The smaller the variance of SNR for a given opportunistic algorithm, lesser is the fluctuation in the throughput along time and along users. Once again the variance is a function of the pdf of the multipath channel and the number of users. Metrics such as the median SNR and variance are useful in analyzing the PFS and MRS algorithms in more detail than just simply looking at the scheduling gains.

IV. Simulations and Discussion

In this section, we will compare the scheduling gains values obtained using EVT based expressions given in the preceding sections with the scheduling gain values obtained through Monte Carlo simulations. Since EVT assumes a large n, our interest is in seeing how well the EVT based expressions for scheduling gains match with the simulated values for realistic values of n.

A fading channel with 10 paths is considered where each tap undergoes Nakagami-m fading with fading parameter being $m_i = m_t$, $\forall i$ ($m_t \geq 0.5$) for all L taps. The power delay profile is exponential, i.e., $a_i = Ae^{-i\delta}$ $i = 0, 1, ...L$ with $\delta = 0.25$. Here, k and θ are computed using (6) and (7) respectively. The parameters of the Gumbel pdf b_n and a_n are computed using (20). The number of users per cell is $n = 30$. Here, $\tilde{G}(n)$ is evaluated by using *Mathematica* to compute (32) for a finite number of terms such that the number of terms used for the computation, m is large enough so that the difference between the spectral efficiency values computed using m terms and $m+1$ terms is less than 0.01% of the spectral efficiency value itself. The infinite sum in (32) converges within 1000 terms ($m = 1000$) for the channel models considered here. The value of g is taken as one in the computation. The values of scheduling gains are tabulated in Table I for a wide range of Nakagami-m parameters. The values computed using the EVT based expressions match very well with the simulation values (columns labelled "Sim"). Since both MRS and PFS have the same scheduling gain over RRS when one is looking at only the small scale fading effect in a single cell scenario we have given only one set of values. Since PFS has the same performance in both small scale fading and composite fading models in the single cell scenario therefore the scheduling gains given here also apply to the scheduling gain of PFS in composite channels. Although the analysis assumes $t_c \to \infty$, it has been observed that a t_c value as low as 100 in simulations gives scheduling gain values comparable to the analytical values.

For the simulated values of scheduling gain in Table II which looks at the scheduling gain of MRS in presence of composite fading, we have dropped $n = 30$ users uniformly in a cell of radius $500m$ (Urban Macro Cell), considering the Urban Macro Pathloss model with a 6 dB lognormal shadow fading component (the simulation parameters are

TABLE I
SCHEDULING GAIN VALUES FOR PFS AND MRS FOR SMALL SCALE FADING CHANNELS (NAKAGAMI DISTRIBUTED) $L = 10$, $\delta = 0.25$

Nakagami Parameters			Gumbel Parameters		$G(n)$		$\tilde{G}(n)$	
m_t	k	θ	b_n	a_n	EVT	Sim	EVT	Sim
0.5	0.872	1.147	3.597	1.111	4.238	4.252	2.768	2.773
0.75	0.953	1.049	3.47	1.037	4.069	4.089	2.684	2.690
1	1	1	3.4	1	3.977	4.001	2.638	2.646
1.25	1.030	0.971	3.36	0.978	3.924	3.927	2.612	2.621
1.5	1.051	0.951	3.336	0.963	3.892	3.91	2.598	2.604
2	1.079	0.927	3.3	0.944	3.845	3.855	2.565	2.580
5	1.133	0.883	3.239	0.910	3.764	3.776	2.544	2.539
10	1.152	0.868	3.218	0.898	3.736	3.745	2.527	2.521

TABLE II
SCHEDULING GAIN VALUES FOR MRS FOR COMPOSITE FADING MODEL WHEN SMALL SCALE FADING IS NAKAGAMI, $L = 10$, $\delta = 0.25$

Nakagami Parameters			Gumbel Parameters		$G(n)$		$\tilde{G}(n)$	
m_t	k	θ	$\tilde{b}_n$	a_n	EVT	Sim	EVT	Sim
0.5	0.872	1.147	15.52	4.86	18.4	12.4	9.65	8.77
0.75	0.953	1.049	15.22	4.64	17.98	12.22	9.58	8.65
1	1	1	15.05	4.54	17.76	12.17	9.55	8.613
1.25	1.03	0.97	14.97	4.47	17.63	12.0	9.52	8.60
1.5	1.051	0.951	14.90	4.43	17.46	12.0	9.51	8.56
2	1.079	0.927	14.81	4.369	17.336	11.92	9.49	8.52
5	1.133	0.883	14.66	4.27	17.125	11.83	9.436	8.466
10	1.152	0.868	14.64	4.24	17.09	11.93	9.442	8.444

commonly used in evaluating the 3GPP LTE standards and more details on them can be found in [24]). The small scale fading is Nakagami-m. For computing the theoretical scheduling gains, the Gumbel pdf in (28) is used. In Table II, $\tilde{b}_n = (c+1)b_n - a_n \log(c+1)$ where a_n and b_n are computed using the generalized-K pdf (11) corresponding to the user with the smallest mean value Ω_i among the 30 users. The value of c is obtained by solving (29). The EVT based scheduling gain $G_{MRS,C}(n)$ is $\tilde{b}_n + \gamma a_n$. Note that in the EVT based analysis it is assumed that the mean received channel powers of the users are non identical but follow a trend with the trend relationship being given by (29). For each and every user drop this assumption may not be strictly accurate over the entire cell. This is the reason why the $G(n)$ values computed using EVT in Table II are higher than the $G(n)$ values obtained in simulation[2]. However, the approach of using a single c value gives theoretical $\tilde{G}(n)$ values which are very close to the simulated values.

To compute the theoretical $\tilde{G}(n)$ values, *Mathematica* is used to evaluate (32) with the input parameters being $g = 1$, a_n and $b_n = \tilde{b}_n$. The infinite sum in (32) converges within 1000 terms ($m = 1000$) for the channel models considered here. Since the rate is usually related logarithmically with SNR the fact that the theoretical values of $\tilde{G}(n)$ are close to the simulated values is significant. Note that the theoretical values of $G(n)$ and $\tilde{G}(n)$ are always higher that the simulated values. This is because the c value used in the analytical computation of the scheduling gains is $c = \sum_{i=1}^{n} w_i \frac{\Omega_{\min}-\Omega_i}{b_i}$ where w_i is a weight given to i^{th} value and is therefore influenced by large values of $\frac{\Omega_{\min}-\Omega_i}{b_i}$. If $\frac{\Omega_{\min}-\Omega_i}{b_i}$ is nearly same for almost all values of i then the theoretical scheduling gains would match exactly with simulation values. However, since the user locations are uniformly distributed random variables, $\frac{\Omega_{\min}-\Omega_i}{b_i}$ need not be similar for all values of i.

Summarizing, the simulated value match exactly with the Gumbel pdf based theoretical values of the scheduling gains in the case where only small scale fading is considered. In the case of the composite channel model, there is a mismatch between the theoretical values of $G(n)$ and the simulation values, however, the theoretical values of $\tilde{G}(n)$ are very close to the simulated values. Though EVT assumes large n, the scheduling gains obtained from the EVT based expressions match with the simulated values even for n values as small as ten.

[2]To further reduce the gap between the EVT based analysis and simulations in the case of the composite channel model, one could divide the cell into zones in a fashion similar to [25] and assume that there is a trend among users in each zone. Instead of one c value for the whole cell this approach would lead to computing a c value for each zone and a corresponding Gumbel pdf for each zone. This in turn would result in the scheduling gain being a function of product of m Gumbel pdfs if there are m zones. However, a detailed analysis of the scheduling gain using the zoning approach is beyond the scope of this paper.

V. CONCLUSIONS

EVT has been used to show that the pdf of the SNR of the users selected by both PFS and MRS is a Gumbel cdf for both the composite channel model and the Nakagami-m/Rayleigh channel model. Though EVT is traditionally used to analyze the maxima of i.i.d. random variables, here we discuss how EVT can be used to analyze maxima of a specific class of independent but non-identical random variables and use this result to find the scheduling gain and spectral efficiency of MRS for the composite channel model. Analytical expressions for scheduling gain derived using EVT have been used to compute the scheduling gains of various Nakagami-m fading channels and composite fading channels and the computed values match very well with the scheduling gain seen in simulations. Furthermore, the moments and order statistics of the Gumbel cdf can be used to study metrics such as the variance of the SNR of selected users, the median of the SNR of selected users for both PFS and MRS and these metrics can give a more detailed understanding of the behavior of these schedulers.

The analysis presented here also holds for TDMA systems with single-tap Rayleigh/Nakagami fading and single-tap composite fading. It is well known that the sum of i.i.d. gamma random variables is a gamma random variable [26], and similarly the cdf of the sum of i.i.d. generalized-K random variables can be approximated by a generalized-K cdf [27]. Therefore, our analysis presented here for SISO systems can be trivially extended to SIMO systems with MRC receiver. Extension of the work to MIMO OFDM systems is a topic for further research.

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APPENDIX A
PROOF THAT THE GENERALIZED-K CDF IS A VON MISES FUNCTION

The pdf of $\tau_i(t)$ in the presence of the composite channel model is the generalized-K pdf which is given by

$$f_{\tau_i(t)}(x) = Ax^{\left(\frac{k+m_s}{2}\right)-1}K_{m_s-k}(l\sqrt{x}), \tag{34}$$

with $x \geq 0$, $k \geq 0.5$, $m_s > 0$ and $A = \frac{2}{\Gamma(k)\Gamma(m_s)}\left(\frac{l}{2}\right)^{k+m_s}$. Differentiating Equation (34) with respect to x and using the relation $K_v^{'}(x) = \frac{-1}{2}\left(K_{v-1}(x) + K_{v+1}(x)\right)$ from [16], we get

$$\begin{aligned} f_{\tau_i(t)}^{'}(x) =& \frac{-Al}{4\sqrt{x}}x^{\left(\frac{k+m_s}{2}\right)-1}\left(K_{m_s-k-1}(l\sqrt{x})\right. \\ &\left. + K_{m_s-k+1}(l\sqrt{x})\right) + \left(\frac{m_s+k}{2} - 1\right)\frac{f_{\tau_i(t)}(x)}{x}. \end{aligned} \tag{35}$$

Using the recurrence relation, $K_v(x) = K_{v+2}(x) - \frac{2(v+1)}{x}K_{v+1}(x)$ from [16], we get

$$K_{m_s-k-1}(l\sqrt{x}) = K_{m_s-k+1}(l\sqrt{x}) - \frac{2(m_s-k)}{l\sqrt{x}}K_{m_s-k}(l\sqrt{x}). \tag{36}$$

Applying Equation (36) in Equation (35) and simplifying further, we obtain

$$f_{\tau_i(t)}^{'}(x) = \frac{f_{\tau_i(t)}(x)}{2x}\left(2(m_s-1) - l\sqrt{x}\frac{K_{m_s-k+1}(l\sqrt{x})}{K_{m_s-k}(l\sqrt{x})}\right). \tag{37}$$

Let $B(x) = \frac{\left(2(m_s-1) - l\sqrt{x}\frac{K_{m_s-k+1}(l\sqrt{x})}{K_{m_s-k}(l\sqrt{x})}\right)}{2x}$. Then $f_{\tau_i(t)}^{'}(x) = B(x)f_{\tau_i(t)}(x)$ and $f_{\tau_i(t)}^2(x)/f_{\tau_i(t)}^{'}(x) = f_{\tau_i(t)}(x)/B(x)$. Hence,

$$\frac{(1-F(x))}{f_{\tau_i(t)}^2(x)/f_{\tau_i(t)}^{'}(x)} = \frac{(1-F(x))}{f_{\tau_i(t)}(x)/B(x)} \tag{38}$$

where $F(x)$ is the generalized-K cdf. Using the property, $\lim_{x\to\infty} K_v(x) = \sqrt{\frac{\pi}{2x}}e^{-x}$, $\lim_{x\to\infty}\frac{f_{\tau_i(t)}(x)}{B(x)}$ can be simplified as

$$\lim_{x\to\infty}\frac{f_{\tau_i(t)}(x)}{B(x)} = \lim_{x\to\infty}\frac{2A\sqrt{\pi}x^{\frac{m_s+k}{2}}e^{-\sqrt{x}}}{\sqrt{2l\sqrt{x}}(2(m_s-1) - l\sqrt{x})} = 0$$

Since $\lim_{x\to\infty} F(x) = 1$, Equation (38) tends to the form $\frac{0}{0}$ as $x \to \infty$. Therefore applying L' Hospital's rule and differentiating the numerator and denominator separately we obtain

$$\lim_{x\to\infty}\frac{f_{\tau_i(t)}^{'}(x)(1-F(x))}{f_{\tau_i(t)}^2(x)} = \lim_{x\to\infty}\frac{-1}{1 - B^{'}(x)/B^2(x)} \tag{39}$$

where $B'(x)$ is the derivative of $B(x)$ w.r.t. x. To show that $\lim_{x\to\infty}\frac{f_{\tau_i(t)}^{'}(x)(1-F(x))}{f_{\tau_i(t)}^2(x)} = -1$, it is sufficient to show that $\lim_{x\to\infty}\frac{B^{'}(x)}{B^2(x)} = 0$. Using the relation $K_v^{'}(x) = \frac{-1}{2}\left(K_{v-1}(x) + K_{v+1}(x)\right)$, $B'(x)$ is given by

$$\begin{aligned} B'(x) = &\frac{1}{8x^2K_{m_s-k}^2(l\sqrt{x})}\Bigg\{(8 - 8m_s + l^2x)K_{m_s-k}^2(l\sqrt{x}) \\ &-l^2xK_{1+m_s-k}(l\sqrt{x})\Big(K_{1+m_s-k}(l\sqrt{x}) + K_{-1+m_s-k}(l\sqrt{x})\Big) \\ &+K_{m_s-k}(l\sqrt{x})\Big(2l\sqrt{x}K_{1+m_s-k}(l\sqrt{x}) + l^2xK_{2+m_s-k}(l\sqrt{x})\Big)\Bigg\}. \end{aligned}$$

Using the property, $\lim_{x\to\infty} K_v(x) = \sqrt{\frac{\pi}{2x}}e^{-x}$ we get

$$\lim_{x\to\infty}\frac{B^{'}(x)}{B^2(x)} = \lim_{x\to\infty}\frac{4 - 4m_s + l\sqrt{x}}{4(-1+m_s)^2 + l^2x - 4(-1+m_s)l\sqrt{x}}. \tag{40}$$

Since the denominator term is of higher order in x than the numerator term, we get

$$\lim_{x\to\infty}\frac{B^{'}(x)}{B^2(x)} = 0. \tag{41}$$

Therefore,

$$\lim_{x\to\infty}\frac{f_{\tau_i(t)}^{'}(x)[1-F(x)]}{f_{\tau_i(t)}^2(x)} = -1. \tag{42}$$

Hence the generalized-K pdf is a von Mises function and it lies in the $\mathcal{MDA}$ of the Gumbel distribution.

APPENDIX B
SPECTRAL EFFICIENCY OF MRS

Using transformation of variable with $y = \frac{x-b_n}{a_n}$ and using the series expansion for e^{-x} we obtain

$$\begin{aligned} C_{MRS} =& \frac{1}{\log 2}\int_{-\frac{b_n}{a_n}}^{\infty}\log(1 + gb_n + ga_ny) \\ &\sum_{m=0}^{\infty}\frac{(-1)^m}{m!}e^{-(m+1)y}dy. \end{aligned} \tag{43}$$

In order to interchange summation and integration, we apply the Lebesgue dominated convergence theorem [28]:

Theorem B.1: Suppose $f_m : M \to \mathbb{R}$ is a sequence of measurable functions, and if we can find another sequence of non-negative measurable functions $d_m : M \to \mathbb{R}$ such that

$|f_m(x)| \leq d_m(x), \forall m$ and almost all x and $\sum_{m=0}^{\infty} d_m(x)$ converges and $\sum_{m=0}^{\infty} \int d_m(x) < \infty$, then

$$\int_M \sum_{m=0}^{\infty} f_m(x)dx = \sum_{m=0}^{\infty} \int_M f_m(x)dx.$$

We can apply Lebesgue theorem by taking $f_m(y) = \frac{(-1)^m}{m!} \log(1 + gb_n + ga_n y)e^{-(m+1)y}$ and $d_m(x) = \log(1 + gb_n + ga_n y)\frac{e^{-(m+1)y}}{m!}$. Applying the ratio test to the series $\sum_{m=0}^{\infty} \log(1 + gb_n + ga_n y)\frac{e^{-(m+1)y}}{m!}$,

$$L = \lim_{m\to\infty} |\frac{d_{m+1}}{d_m}| = \lim_{m\to\infty} \frac{e^{-y}}{m+1} = 0.$$

Since $L < 1$, the series converges. In addition, we also require that $\sum_{m=0}^{\infty} \int_{\frac{-b_n}{a_n}}^{\infty} \log(1 + gb_n + ga_n y)\frac{e^{-(m+1)y}}{m!}dy < \infty$. The integral in the summation can be written as:

$$\int_{\frac{-b_n}{a_n}}^{\infty} \log(1 + gb_n + ga_n y)\frac{e^{-(m+1)y}}{m!}dy = \frac{e^{\frac{(m+1)(1+gb_n)}{ga_n}}}{m!} \int_1^{\infty} \log(z)e^{-\frac{(m+1)z}{ga_n}}dz. \quad (44)$$

Applying transformation of variables with $z = 1+gb_n+ga_n y$ and using formula 4.331.2 in [29], we can show that

$$\frac{e^{\frac{(m+1)(1+gb_n)}{ga_n}}}{ga_n m!} \int_1^{\infty} \log(z)e^{-\frac{(m+1)z}{ga_n}}dz = -\frac{e^{\frac{(m+1)(1+gb_n)}{ga_n}}}{(m+1)!} Ei(-\frac{m+1}{ga_n}) \quad (45)$$

where $Ei(\cdot)$ is the exponential integral [29]. The above equation can be further simplified and written as

$$\int_{\frac{-b_n}{a_n}}^{\infty} \log(1 + gb_n + ga_n y)\frac{e^{-(m+1)y}}{m!}dy = \frac{e^{\frac{(m+1)(1+gb_n)}{ga_n}}}{(m+1)!}\Gamma(0, \frac{m+1}{ga_n}) \quad (46)$$

where $\Gamma(0, x)$ is the upper incomplete Gamma function. Hence the m^{th} term, s_m of $\sum_{m=0}^{\infty} \int_{\frac{-b_n}{a_n}}^{\infty} \log(1 + gb_n + ga_n y)\frac{e^{-(m+1)y}}{m!}dy$ given by Equation (46) is always non-negative, and the series converges if $\lim_{m\to\infty} \frac{s_{m+1}}{s_m} < 1$ [28]. Denoting $c = \frac{1+gb_n}{ga_n}$ and simplifying $\frac{s_{m+1}}{s_m}$ we get

$$\lim_{m\to\infty} \frac{s_{m+1}}{s_m} = \lim_{m\to\infty} \frac{e^c}{(m+2)} \frac{\Gamma(0, \frac{m+2}{ga_n})}{\Gamma(0, \frac{m+1}{ga_n})}. \quad (47)$$

The 2^{nd} term in the above equation $x = \frac{\Gamma(0, \frac{m+2}{ga_n})}{\Gamma(0, \frac{m+1}{ga_n})}$ is less than one since $\Gamma(0, \frac{m+1}{ga_n}) > \Gamma(0, \frac{m+2}{ga_n})$. Hence

$$\lim_{m\to\infty} \frac{s_{m+1}}{s_m} = \lim_{m\to\infty} \frac{xe^c}{m+2} = 0 < 1 \quad (48)$$

Hence, the series converges, and we can interchange integration and summation in Equation (43). Interchanging integration and summation, one obtains the expression for spectral efficiency as:

$$C_{MRS} = \frac{1}{\log 2} \sum_{m=0}^{\infty} \frac{(-1)^m}{(m+1)!} e^{\frac{(m+1)(1+gb_n)}{ga_n}} \Gamma(0, \frac{m+1}{ga_n}). \quad (49)$$

Equation (49) is an alternating series of the form $\sum_{m=1}^{\infty}(-1)^m s_m$ with $s_m > 0$. Since $\sum_{m=1}^{\infty} s_m$ converges, the series $\sum_{m=1}^{\infty}(-1)^m s_m$ converges absolutely [28] and hence C_{MRS} converges. Therefore, C_{MRS} can be computed by evaluating the infinite sum in (32) for finite number of terms.

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